

infrarails - Compliance Report

EU AI Act Article 12 · NIST AI RMF · ISO/IEC 42001
2026-05-07 06:16:18 UTC · 14 findings

5	1	0	6	2
failed	warnings	inconclusive	passed	skipped

FAILFailures - fix these5

FAIL S-12.1.2b

No `aws_s3_bucket_lifecycle_configuration` found for log bucket "my-unprotected-large-data-bucket". Without a lifecycle policy, retention is governed only by manual deletion - there is no enforced minimum.

! Add an `aws_s3_bucket_lifecycle_configuration` for "my-unprotected-large-data-bucket" with an expiration rule of ≥ 180 days (recommended: 365). Why: Article 12 requires logs retained for an "appropriate period to the intended purpose" - no specific number is named, but bias drift, hallucinated decisions, and downstream-deployer audits routinely surface months after the event. Sub-180-day retention undermines post-market monitoring (Article 72) and incident investigation; 365 days is the typical floor for production AI.

EU AI ACT Article 12(1) NIST AI RMF MANAGE 4.1 MANAGE 4.3 MEASURE 3.2 ISO/IEC 42001 A.6.2.8 A.6.2.6

FAIL S-12.1.2b /home/vbalaji/ai-compliance-scanner/infrarails/test/fixtures/non-compliant/main.tf:18

Lifecycle retention for bucket "my-ai-log-bucket" is 30 days, below the 180-day floor infrarails applies for high-risk AI logging.

! Increase `expiration.days` to ≥ 180 (recommended: 365). Why: Article 12 requires logs retained for an "appropriate period to the intended purpose" - no specific number is named, but bias drift, hallucinated decisions, and downstream-deployer audits routinely surface months after the event. Sub-180-day retention undermines post-market monitoring (Article 72) and incident investigation; 365 days is the typical floor for production AI.

EU AI ACT Article 12(1) NIST AI RMF MANAGE 4.1 MANAGE 4.3 MEASURE 3.2 ISO/IEC 42001 A.6.2.8 A.6.2.6

FAIL S-12.x.1

Log bucket "my-ai-log-bucket" has neither versioning nor Object Lock enabled - a same-key PUT or a `DeleteObject` call will silently overwrite or remove log entries with no recoverable history.

! Add `aws_s3_bucket_versioning` with `versioning_configuration.status = "Enabled"`, or `aws_s3_bucket_object_lock_configuration`. Why: Article 12 requires log integrity sufficient for downstream auditability. Without versioning, an attacker (or a script bug) can overwrite a log object using the same key and erase forensic evidence - by the time you notice, the original is gone. Best practice for high-risk AI is Object Lock in COMPLIANCE mode (immutable for the retention period); versioning + a bucket policy that denies `s3:DeleteObject` is acceptable for lower-risk systems.

EU AI ACT Article 12 NIST AI RMF MEASURE 2.7 MANAGE 4.3 ISO/IEC 42001 A.6.2.8 A.6.2.4

FAIL S-12.x.2a /home/vbalaji/ai-compliance-scanner/infrarails/test/fixtures/non-compliant/main.tf:8

Log bucket "my-ai-log-bucket" uses "AES256" encryption instead of KMS - log access is encrypted at rest but is not auditable per-call.

! Set `sse_algorithm` to "aws:kms" or "aws:kms:dsse" in the encryption configuration. Why: AES256 (SSE-S3) does not emit CloudTrail Decrypt events, so reads of AI logs are invisible to your audit trail. KMS Decrypt events name the principal who read the data - Article 12 requires this level of traceability for downstream-deployer access.

EU AI ACT Article 12 NIST AI RMF MEASURE 2.7 MEASURE 2.10 ISO/IEC 42001 A.6.2.8 A.7.4

FAIL S-12.x.4 /home/vbalaji/ai-compliance-scanner/infrarails/test/fixtures/non-compliant/main.tf:50

CloudTrail "main" has `enable_logging` set to false - no control-plane events are being captured.

! Set `enable_logging = true` on the `aws_cloudtrail` resource.

EU AI ACT Article 12 NIST AI RMF MANAGE 4.1 GOVERN 1.4 MEASURE 2.7 ISO/IEC 42001 A.6.2.8 A.3.3

WARN

Warnings

1

WARN

S-12.1.2a/home/vbalaji/ai-compliance-scanner/infrarails/test/fixtures/non-compliant/main.tf:31

CloudWatch log group "/aws/bedrock/invoke-log" retention is 7 days, below the 180-day floor infrarails applies for high-risk AI logging.

! Increase retention_in_days to >= 180 (recommended: 365), or confirm retention is satisfied by a forwarder destination / central log-archive account. No CloudWatch subscription filter was found in the scanned Terraform, but forwarders are commonly owned by a separate platform repo (Datadog/Splunk forwarder Lambda, central log-archive account, auto-subscription Lambda). If logs are forwarded out-of-repo, verify retention at the destination. If they are not, this is a real compliance gap. Why retention matters: Article 12 requires logs retained for an "appropriate period to the intended purpose" - no specific number is named, but bias drift, hallucinated decisions, and downstream-deployer audits routinely surface months after the event. Sub-180-day retention undermines post-market monitoring (Article 72) and incident investigation; 365 days is the typical floor for production AI.

EU AI ACT Article 12(1) NIST AI RMF MANAGE 4.1 MANAGE 4.3 MEASURE 3.2 ISO/IEC 42001 A.6.2.8 A.6.2.6

PASS

Passing

6

PASS

/home/vbalaji/ai-compliance-scanner/infrarails/test/fixtures/non-compliant/cw-large-data-missing-lifecycle/main.tf:64

Bedrock invocation logging is configured (main).

EU AI ACT Article 12(1) NIST AI RMF GOVERN 1.4 MEASURE 2.7 MANAGE 4.1 ISO/IEC 42001 A.6.2.8 A.6.2.6

PASS

S-12.1.1 /home/vbalaji/ai-compliance-scanner/infrarails/test/fixtures/non-compliant/main.tf:36

Bedrock invocation logging is configured (main).

EU AI ACT Article 12(1) NIST AI RMF GOVERN 1.4 MEASURE 2.7 MANAGE 4.1 ISO/IEC 42001 A.6.2.8 A.6.2.6

PASS

/home/vbalaji/ai-compliance-scanner/infrarails/test/fixtures/non-compliant/cw-large-data-missing-lifecycle/main.tf:59

CloudWatch log group "/aws/bedrock/regression-log" retention is 365 days.

EU AI ACT Article 12(1) NIST AI RMF MANAGE 4.1 MANAGE 4.3 MEASURE 3.2 ISO/IEC 42001 A.6.2.8 A.6.2.6

PASS

/home/vbalaji/ai-compliance-scanner/infrarails/test/fixtures/non-compliant/cw-large-data-missing-lifecycle/main.tf:4

Log bucket "my-unprotected-large-data-bucket" has versioning enabled.

EU AI ACT Article 12 NIST AI RMF MEASURE 2.7 MANAGE 4.3 ISO/IEC 42001 A.6.2.8 A.6.2.4

PASS

/home/vbalaji/ai-compliance-scanner/infrarails/test/fixtures/non-compliant/cw-large-data-missing-lifecycle/main.tf:12

Log bucket "my-unprotected-large-data-bucket" uses aws:kms encryption.

EU AI ACT Article 12 NIST AI RMF MEASURE 2.7 MEASURE 2.10 ISO/IEC 42001 A.6.2.8 A.7.4

PASS

/home/vbalaji/ai-compliance-scanner/infrarails/test/fixtures/non-compliant/cw-large-data-missing-lifecycle/main.tf:81

CloudTrail "main" is configured with logging enabled.

EU AI ACT Article 12 NIST AI RMF MANAGE 4.1 GOVERN 1.4 MEASURE 2.7 ISO/IEC 42001 A.6.2.8 A.3.3

SKIP

Skipped (not applicable)

2

SKIP

S-9.x.1

No Bedrock Agents detected. Guardrail attachment check skipped.

EU AI ACT Article 9 NIST AI RMF MEASURE 2.6 MAP 5.1 GOVERN 1.4 ISO/IEC 42001 A.6.1.2 A.6.2.4 A.6.2.5

SKIP

S-9.x.2

No Bedrock workload detected (no Bedrock resources, IAM grants, VPC endpoints, or data sources). Guardrail-presence check skipped.

EU AI ACT Article 9 NIST AI RMF MEASURE 2.6 MAP 5.1 GOVERN 1.4 ISO/IEC 42001 A.6.1.2 A.6.2.4 A.6.2.5

Disclaimer

This report reflects the findings of an automated static analysis of your AWS AI infrastructure configuration against selected controls from the EU AI Act, NIST AI RMF, and ISO/IEC 42001. A passing result indicates that the scanned Terraform configuration satisfies the specific infrastructure-layer prerequisite checked - it does not constitute compliance with any of these frameworks, nor does it substitute for a formal audit, certification, or conformity assessment conducted by an accredited body. Compliance with the EU AI Act, NIST AI RMF, and ISO/IEC 42001 requires organisational, procedural, and governance measures that are outside the scope of infrastructure scanning. This report should be treated as a pre-audit readiness input, not an attestation of conformance.